NanoGT Match Report: Phenylketonuria

Disease: Phenylketonuria (ORPHA:716)
Primary gene: PAH
Gene CDS: 1353 bp
Inheritance: Autosomal recessive
Target tissues: liver, CNS

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	BMN 307	AAV5	7.6/10	[GREEN] High	phase2
2	OAV101-IT	AAV9	7.3/10	[YELLOW] Medium	approved
3	Zolgensma	AAV9	7.3/10	[YELLOW] Medium	approved
4	Hemgenix	AAV5	7.1/10	[YELLOW] Medium	approved
5	Roctavian	AAV5	7.1/10	[YELLOW] Medium	approved

Match #1: BMN 307

Precedent disease: Phenylketonuria
Vector: AAV5
Tissue target: liver
Composite score: 7.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**7.61**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #2: OAV101-IT

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/spinal cord
Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**7.28**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #3: Zolgensma

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/motor neuron
Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**7.28**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #4: Hemgenix

Precedent disease: Hemophilia B
Vector: AAV5
Tissue target: liver
Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**7.11**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #5: Roctavian

Precedent disease: Hemophilia A
Vector: AAV5
Tissue target: liver
Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**7.11**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs